Md. Abul Hasan

Senior Staff Software Engineer

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13+	1M+	100K+	5
Years shipping production software	Users on the public Arkivportalen portal I own	EU users on the fintech platform I led	Codebases modernized — 4× Angular 14→21, 1× React

I architect and ship frontends for nation-scale platforms, and work across the stack when delivery requires it. At senior staff level I focus on architecture that survives scale, delivery pressure and team changes — writing production code, driving technical direction and mentoring engineers. I run an agentic development workflow end-to-end: an agent pipeline carries a Jira ticket through analysis, design spec, plan, implementation, code review and security review, while I direct the architecture and own every merge. Every role in my career has run on Agile/Scrum — sprints, retros and regular releases.

EXPERIENCE

Senior Staff Software Engineer · Cefalo — Aug 2022 – Present

Project: Archive management system for the Norwegian national archives — 5 modules, microservice-backed

- Own the frontend of **all 5 modules**, plus the Arkivportalen portal and requisition backends end-to-end — leading the team day-to-day, with oversight of backend work on the remaining modules
- Led the **Angular 14→21 migration across four codebases** and one React migration — pilot repo first, then the rest in overlap; ~2 months with a team cut from 10 to 5. Angular Material was deliberately pinned at v14 during the migration to avoid the MDC breaking changes; that pin has since been lifted and the clients now run Material 21
- **Re-architected the Kafka CDC pipeline** (Debezium, Kafka Connect) for reliability and rolled the new architecture out across four modules, keeping Elasticsearch and cross-module shared data in sync with PostgreSQL/MongoDB
- Built a **digital-archive middleware in Python** — pulls portal data into its own PostgreSQL and reconciles publish status against an external digital-media system, with per-item checkpointing and retries; the portal never calls the external system
- Built the dynamic permission engine and Elasticsearch-backed search; implemented WebSocket live notifications; Google Analytics integration and UI improvements on the portal
- Set up self-hosted Bitbucket Pipelines runners on Azure VMs for CI
- Introduced the **agentic delivery workflow** now used by the whole team — see the next section
- **Outcome:** production system serving 10,000+ archivists and a public portal with 1M+ users; migration roadmap delivered with a 50% leaner team

Stack: Angular 21, React 19, TypeScript, Kotlin, Java 25, Spring Boot, Python, PostgreSQL, MongoDB, Elasticsearch, Apache Kafka, Debezium, ActiveMQ, Keycloak, Docker, Azure

Principal Software Engineer · SELISE Digital Platforms — 2018 – Jul 2022

Project: Fintech platform connecting IFAs and their customers, EU — multi-tenant, microservice-backed advisor & customer portal

- Owned core frontend architecture for a platform serving **100,000+ users** across the EU on 4 tenant builds; led the 12-member frontend team within a ~20-member product team
- Built the company-wide Angular frontend architecture (SSR-capable) used by all SELISE teams — teams published shared components, and common packages (auth, SSO, SignalR-integrated notifications, dynamic datatable, dialog) shipped on an internal npm registry
- Delivered SSO (OIDC), multi-tenancy via a shared component library with separate per-tenant builds, PWA and push notifications
- Real-time delivery via SignalR + RabbitMQ; data-intensive dashboards with Chart.js and AG Grid
- Migrated 4 projects from Angular 2 to 12

Stack: Angular, TypeScript, .NET Core, C#, MongoDB, Redis, RabbitMQ, SignalR, JWT, PWA

Earlier Career — 2013 – 2017

- **Senior Software Engineer** — Wealth & insurance portal (2017): internal/external insurance portal; AngularJS + .NET Core microservices, SignalR real-time
- **Software Engineer** — E-services for a Swiss IT firm (2016): reservation platform, social-feed aggregation, public statistics dashboards
- **Developer** — Distribution ERP, Bangladesh (2013–2015): end-to-end supply-chain ERP — inventory, RMA, sales, POS, accounts, PMS

AI & AGENTIC DEVELOPMENT

The pipeline

- Seven stages: read the Jira ticket, analyze the requirement, design spec, implementation plan, implement, code review, security review
- Runs on **Claude Code**. Generic stages use the Superpowers skill library; project-specific stages use team-written skills
- Spec and plan publish to Confluence and back to the Jira ticket
- I own every merge — no agent output reaches a shared branch without human review

Guardrails I built

- Wrote **all seven Claude Code hooks** in the repository
- Block pushes, block commits on shared branches, block no-verify bypasses
- Shipped DB migrations made immutable once they reach the integration branch
- Credential files blocked from reads and commits; tool permission rules (read-only Figma, no Jira deletes)

Team enablement

- Wrote the **CLAUDE.md** conventions for the monorepo and nine module directories
- Wrote the team's Jira ticket-creation skill (mirroring the Rovo Task Creator) and the i18n authoring rule
- Set up the MCP server configuration: Jira, Confluence, Figma, Context7
- Took part in a two-month team AI pilot; workflow published as a public case study

Other AI work

- i18n tooling engine assigning stable identifiers to translation strings — removed a recurring class of merge conflict
- AI-assisted test generation for Angular unit tests and Playwright E2E specs
- Design-to-code from Figma through MCP
- Commit throughput rose from ~44/month to 200+/month after adoption

ARCHITECTURE CASE STUDIES

Migrating 5 codebases to Angular 21 with half the team — sequencing and risk management
ahasan09.github.io/case-studies/angular-migration.html

An agentic workflow: from Jira ticket to reviewed PR — agent pipeline with a human owning every merge
ahasan09.github.io/case-studies/agentic-workflow.html

CDC with Kafka: keeping Elasticsearch in sync — why change data capture beat dual writes
ahasan09.github.io/case-studies/kafka-cdc-search.html

A pull-only middleware: publish status behind a rate-limited API — per-item checkpointing at 1M-user scale
ahasan09.github.io/case-studies/archive-middleware.html

A multi-tenant fintech platform for 100K EU users — SSO, tenancy isolation, PWA, real-time
ahasan09.github.io/case-studies/fintech-platform.html

STACK

Frontend	Angular, React, TypeScript, JavaScript, RxJS, Redux, Angular Material, PrimeNG, PrimeReact, Bootstrap, PWA, SSR
Backend	Kotlin, Java, Spring Boot, .NET Core, C#, Node.js, NestJS, Express, Python, REST, GraphQL, JWT, OAuth2, Keycloak, SSO
Data & messaging	PostgreSQL, MongoDB, MSSQL, Redis, Elasticsearch, Apache Kafka, Debezium CDC, RabbitMQ, ActiveMQ, SignalR, WebSocket
Cloud & delivery	Docker, Bitbucket Pipelines, Microsoft Azure, AWS, Azure DevOps, Jenkins, GitHub Actions, Application Insights, Azure Blob Storage, Grafana/Loki
AI & agentic	Claude Code, custom agent skills, Claude Code hooks, MCP servers (Jira, Confluence, Figma, Context7), agent-assisted code & security review, prompt and context engineering
Quality & practice	Playwright, Jest, Jasmine, Karma, SonarQube, TDD, DDD, SOLID, Design Patterns, Clean Architecture, Agile/Scrum

EDUCATION & CERTIFICATION

- M.Sc. in Computer Science & Engineering — Jahangirnagar University
- B.Sc. in Computer Science & Engineering — Patuakhali Science and Technology University
- C# .NET Certification — BASIS (2013)
- Claude Certified Architect – Foundations — Anthropic (Nov 2026) *— verify on Credly*

Recommendations from managers and peers: linkedin.com/in/ahasan09 · Full case studies with architecture diagrams: ahasan09.github.io